M2S-02: Step 2 — Strategize: Define Your Migration Approach

> **Series:** M2S — Managed to SaaS Migration | **Notebook:** 2 of 9 | **Phase:** Plan | **Step:** Strategize | **Created:** March 2026 | **Last Updated:** 07/24/2026

With your discovery complete, it's time to turn inventory into action. This notebook helps you select a migration approach, sequence your operations, assess risks, and build a timeline that earns stakeholder confidence.

> **M2S Migration Journey — 3 Phases / 9 Steps**
>
> **Plan:** 1. Discover | **2. Strategize** | 3. Design
>
> **Upgrade:** 4. Prepare | 5. Execute | 6. Integrate
>
> **Run:** 7. Enable | 8. Expand | 9. Optimize

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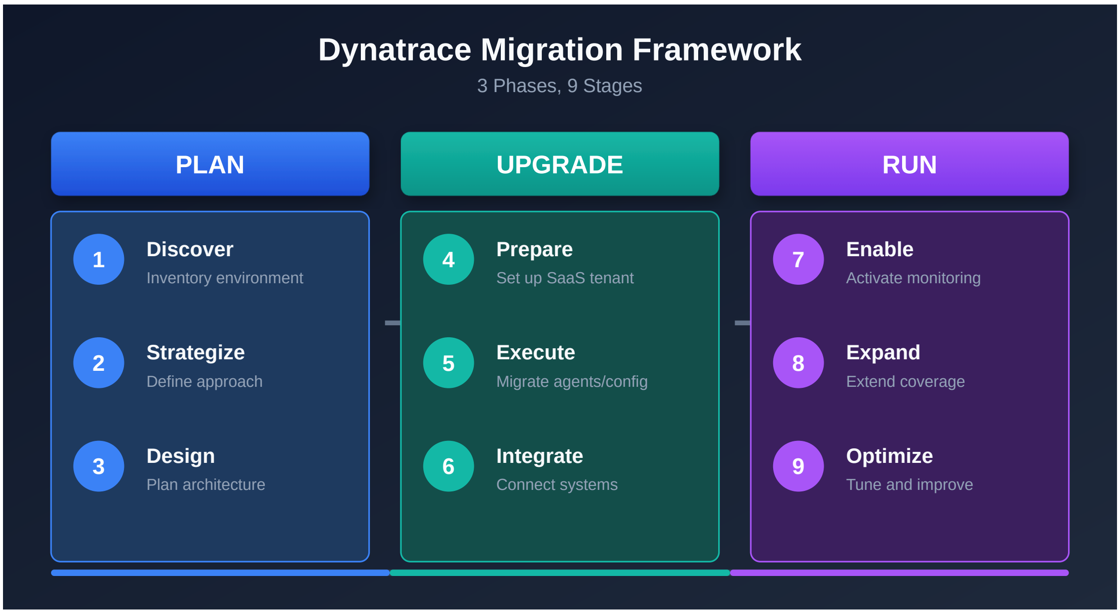
Prerequisites

Requirement	Description
Step 1 Complete	Discovery inventory from M2S-01 (host counts, agent versions, ActiveGate topology, integrations)
Stakeholder Access	Ability to engage infrastructure, security, and application teams
Dynatrace Account Team	Contact established for licensing and migration support
Risk Tolerance Understood	Organizational appetite for maintenance windows and parallel operation



1. The Three-Phase Framework

The Managed-to-SaaS migration follows a proven three-phase, nine-step framework. Each phase builds on the previous one, and skipping steps leads to preventable failures.

Phase Overview

Phase	Steps	Focus
Plan	1. Discover → 2. Strategize → 3. Design	Understand what you have, decide how to move it, design the target
Upgrade	4. Prepare → 5. Execute → 6. Integrate	Set up SaaS, migrate agents and config, reconnect integrations
Run	7. Enable → 8. Expand → 9. Optimize	Enable teams, adopt new capabilities, tune for value

What Each Step Delivers

Step	Name	Key Deliverable
1	**Discover**	Complete environment inventory (hosts, agents, configs, integrations)
2	**Strategize**	Migration approach, timeline, risk assessment, success criteria
3	**Design**	Target SaaS architecture, network topology, IAM model
4	**Prepare**	SaaS tenant provisioned, ActiveGates deployed, network validated
5	**Execute**	Configurations migrated, OneAgents redirected, data flowing
6	**Integrate**	Webhooks, CI/CD, ITSM, and third-party tools reconnected
7	**Enable**	User communication, persona-based training, documentation, and support channels
8	**Expand**	Grail, Notebooks, OpenPipeline, Workflows, and Dynatrace Assist adopted
9	**Optimize**	Alert tuning, cost management, query performance, feature adoption

Time Investment by Phase

Phase	Typical Duration	Effort Level
Plan	2-4 weeks	Medium — analysis and decision-making
Upgrade	2-6 weeks	High — technical execution

Phase	Typical Duration	Effort Level
Run	2-4 weeks	Medium — enablement and tuning

> **Note:** Duration varies significantly based on environment size, complexity, and organizational readiness. A 200-host environment can complete in 4 weeks. A 5,000-host multi-region deployment may take 12 weeks.

2. Migration Approach Selection

The single most consequential decision in your migration is the approach. Choose based on environment size, risk tolerance, and operational constraints.

Three Approaches

Approach	When to Use	Pros	Cons
Big Bang	< 500 hosts, simple environment	Fast, decisive, clean cutover	Higher risk, requires extensive prep
Phased by Environment	500-2,000 hosts	Lower risk, lessons learned per wave	Longer timeline, dual-run costs
Phased by Region/App	> 2,000 hosts or complex integrations	Lowest risk, maximum flexibility	Longest timeline, complex coordination

Big Bang Migration

All agents and configurations migrate in a single maintenance window.

Factor	Detail
Typical window	4-8 hours
Best for	Smaller environments with flexible maintenance windows
Requires	Complete preparation, tested rollback procedure
Risk	Higher — all-or-nothing in a single window

Phased by Environment

Migrate in waves: Dev → Staging → Production.

Factor	Detail
Typical duration	3-6 weeks (1-2 weeks per wave)
Best for	Mid-size environments with distinct environment tiers
Requires	Parallel operation budget, wave-by-wave validation
Risk	Medium — each wave validates the next

Phased by Region or Application

Migrate by geography (EMEA → APAC → Americas) or by application criticality.

Factor	Detail
Typical duration	6-12 weeks
Best for	Large, complex environments with regional or app-team ownership
Requires	Complex coordination, extended parallel operation
Risk	Lowest — isolated blast radius per wave

Decision Matrix

Factor	Big Bang	Phased by Env	Phased by Region/App
Hosts < 500	Recommended	Optional	Unnecessary
Hosts 500-2,000	Possible	Recommended	Optional
Hosts > 2,000	Not recommended	Possible	Recommended
Simple integrations	Recommended	Optional	Unnecessary
Complex integrations	Risky	Recommended	Recommended
Short timeline required	Fastest	Moderate	Longest

Factor	Big Bang	Phased by Env	Phased by Region/App
Risk-averse organization	Higher risk	Lower risk	Lowest risk

Cloud-Native and Serverless Estates

Host count is the wrong yardstick for a Cloud Run / GKE / OTLP-heavy estate. A shop with thousands of Cloud Run services and a handful of GKE clusters may report only a few residual VMs, yet its migration is far from "small." Size the approach on the axes that actually drive coordination and risk in a cloud-native estate:

Cloud-native axis	Big Bang	Phased by Env	Phased by Cluster/Service
Cloud Run / serverless services	< ~20 services, single team	Dozens across a few teams	Hundreds of services or many owning teams
GKE / Kubernetes clusters	1 cluster	2–3 clusters	4+ clusters or multi-region
Log volume & pipeline complexity	Low, few sources	Moderate, some OpenPipeline routing	High volume, many sources / bucket-routing rules
Redirect blast radius	One redeploy window	Per-environment redeploys	Per-cluster / per-service-team redeploys

Map these onto the same risk profiles as the host-count bands above: a single GKE cluster with modest log volume behaves like a **Big Bang** candidate, while many clusters or high log volume push you toward **Phased by Cluster/Service** for an isolated blast radius. Two cloud-native specifics change the mechanics:

- **The redirect is a redeploy, not an `oneagentctl` reconfigure** — a new Cloud Run revision or rolled GKE pods. See **M2S-05: Step 5 — Execute** for the per-surface redirect matrix.
- **ActiveGate footprint is minimal** — cloud-native estates route far less through ActiveGates (forwarder path, private synthetics, and residual VMs only). See **M2S-03: Step 3 — Design** for cloud-native AG sizing.

3. Order of Operations

Regardless of which approach you choose, the migration must follow this exact 11-step sequence. Steps cannot be reordered — each depends on the previous one completing successfully. The rightmost column shows which notebook covers each operation in detail.

Sequence	Activity	Why This Order	Covered In
1	**Assess** — Inventory current environment	Cannot plan what you don't know	Step 4: Prepare
2	**Provision** — SaaS tenant and access (SSO)	Target environment must exist before anything moves	Step 4: Prepare
3	**Install** — New ActiveGates in parallel with old ActiveGates	Agents need routing infrastructure before redirect	Step 4: Prepare
4	**Migrate** — Configuration and integrations	Settings must be in place before agents report to SaaS	Step 5: Execute
5	**Rebuild** — Dashboards, zones, alerts	Operational visibility must be ready before cutover	Step 5: Execute
6	**Redirect** — OneAgents to SaaS	The actual migration moment — agents begin reporting to SaaS	Step 5: Execute
7	**Reconnect** — Integrations and extensions	Restore external system connections	Step 6: Integrate
8	**Migrate** — Any remaining configuration and integrations	Catch items missed in first pass or dependent on agent data	Step 6: Integrate
9	**Validate** — Data flow and performance	Confirm everything works before declaring success	Step 9: Optimize
10	**Cutover** — Full switch to SaaS	Formal declaration that SaaS is the primary platform	Step 9: Optimize
11	**Decommission** — Managed environment	Only after full validation and parallel observation period	Step 9: Optimize

> **Important:** Steps 4 and 5 are where the SaaS Upgrade Assistant does the heavy lifting. Export from Managed, upload to SaaS, review, and deploy — most configurations migrate automatically.

Critical Dependencies

Dependency	Impact if Missed
ActiveGates before OneAgents	Agents have nowhere to route — data loss
Config before agent redirect	No alerting, no dashboards during cutover
SSO before user access	Teams locked out of new SaaS tenant
Firewall rules before anything	All traffic blocked — complete failure

4. Migration Considerations and Dependencies

These are the non-obvious factors that derail migrations when overlooked.

Licensing and Contracts

Consideration	Detail
Dual licensing	You will run both Managed and SaaS in parallel during migration — coordinate with your Dynatrace account team early
Contract alignment	SaaS licensing model differs from Managed — DPS (Dynatrace Intelligence Processing Units) vs. host-based
Timing	Begin licensing discussions at least 4 weeks before migration start

Data Continuity

Consideration	Detail
Historic data cannot be migrated	Managed data stays in Managed — plan for a baseline period in SaaS

Consideration	Detail
Parallel observation	Run both environments for 2-4 weeks to build SaaS baselines
Data gap planning	Brief gap during OneAgent redirect is unavoidable — minimize with maintenance window

Infrastructure

Consideration	Detail
New ActiveGates required	SaaS requires Environment ActiveGates — cannot reuse Managed Cluster ActiveGates
Network zones	Must be recreated in SaaS — plan ActiveGate placement per zone
Firewall rules	New outbound rules to `*.live.dynatrace.com` and `*.apps.dynatrace.com` on port 443
OneAgent version compatibility	Dynatrace supports OneAgent versions for 9 months (Standard) / 12 months (Enterprise) — verify your oldest agents are within support

Configuration

Consideration	Detail
Environment rebuild required	SaaS is a new environment — settings, dashboards, and alerts must be migrated or recreated
Credential Vault	Manual migration — secrets cannot be exported from Managed
Synthetic private locations	Must be recreated with new ActiveGates in SaaS
Custom extensions	Extensions 2.0 required — legacy extensions need conversion
Configuration freeze	Freeze Managed changes during migration to prevent drift

Security and Compliance

Consideration	Detail
Security approvals	Firewall change requests may require weeks of lead time
SSO/SAML configuration	IdP must sign the **entire SAML message**, not just the assertion
SSO federation limits	SaaS supports **SAML 2.0 only** (plus SCIM provisioning) — OIDC federation and direct LDAP are not available; confirm your IdP can federate via SAML
Outbound email (SMTP)	No custom SMTP on SaaS — notifications are sent by Dynatrace; route to an internal relay via a webhook or Workflow email action if required
API token rotation	New tokens needed for SaaS — update all automation and integrations
Data residency	Confirm SaaS tenant region meets compliance requirements

Execution Readiness

Consideration	Detail
Rollback procedure	Document and test on one host before full migration
Maintenance window	Coordinate with all application teams
Communication plan	Stakeholders must know what's happening and when
Version alignment	Align Managed cluster and SaaS to the same major version for SaaS Upgrade Assistant

Use Your Discovery Data to Identify Dependencies

The queries from Step 1 (Discover) provide the data you need to assess these dependencies. If you haven't run them yet, go back to **M2S-01** and complete the discovery first.

Once your agents are reporting to SaaS, use these DQL queries to validate coverage and identify gaps.

```
// Check OneAgent version distribution — agents outside the 9-month (Standard)
/ 12-month (Enterprise) support window need upgrading before migration
fetch dt.entity.host
| fieldsAdd version = installerVersion
| summarize hostCount = count(), by:{version}
| sort hostCount desc

// No Smartscape equivalent: installerVersion (OneAgent version) is not a
Smartscape node
// field, so this version distribution stays on the classic entity store.
```

```
// Count monitored entities by type — confirms your discovery inventory
matches what SaaS sees after redirect
fetch dt.entity.host | summarize hosts = count()
| append [fetch dt.entity.service | summarize services = count()]
| append [fetch dt.entity.application | summarize applications = count()]
| append [fetch dt.entity.process_group | summarize process_groups = count()]

// Keep classic: this is a completeness inventory spanning types not all
present on Grail
// Smartscape (e.g. application). Smartscape would count live topology, not
the full
// monitored inventory — the wrong basis for a migration-completeness check.
```

```
// Verify ActiveGate connectivity — all Environment ActiveGates should be
reporting
fetch dt.entity.active_gate
| fieldsAdd entity.name, networkZone
| sort entity.name asc

// Note: smartscapeNodes ACTIVE_GATE is not yet available on Grail
// Continue using fetch dt.entity.active_gate until Smartscape coverage
expands
```

5. Risk Assessment

Every migration carries risk. The goal is not to eliminate risk but to identify, quantify, and mitigate it before execution begins.

Risk Register

Risk	Impact	Likelihood	Mitigation
Data gaps during cutover	High	Medium	Plan maintenance window; use parallel install to minimize gap to < 15 minutes
Configuration drift	Medium	High	Freeze Managed changes during migration; use SaaS Upgrade Assistant for snapshot export
Integration failures	High	Medium	Test all webhooks and APIs in SaaS before cutover; verify endpoints and tokens
Rollback needed	Medium	Low	Document rollback procedure; test on one host first; keep Managed running during parallel period
Licensing overlap costs	Low	High	Coordinate with Dynatrace account team early; align contract dates with migration timeline
SSO/SAML misconfiguration	High	Medium	Test IdP integration before any user access; verify full SAML message signing
Network connectivity blocked	High	Medium	Submit firewall change requests 4+ weeks early; test connectivity before migration day
Unsupported OneAgent versions	Medium	Low	Audit agent versions in discovery; upgrade agents outside support window before redirect

Risk Scoring

Impact Level	Definition
High	Migration failure, data loss, or monitoring gap > 1 hour
Medium	Partial functionality loss or delayed timeline
Low	Inconvenience or minor cost impact

> **Tip:** Assign an owner to each risk. Unowned risks are unmitigated risks.

6. Defining Success Criteria

Define measurable success criteria before migration starts. These criteria determine when the migration is "done" and when you can decommission Managed.

Target Metrics

Metric	Target	How to Measure
Host coverage	100% of discovered hosts reporting	Compare DQL host count to discovery inventory
Service discovery	100% of known services detected	Compare DQL service count to discovery inventory
Data gaps	< 15 minutes during cutover	Review data availability in SaaS after redirect
Alert delivery	100% of critical alerts firing correctly	Trigger test alerts; verify notification channels
Dashboard availability	100% of migrated dashboards rendering	Spot-check each dashboard in SaaS
Integration success	100% of external integrations operational	Test each webhook, API, and ITSM connection
User access	All users can authenticate via SSO	Verify SSO login for each role/group
Baseline established	Dynatrace Intelligence baseline period complete (2-4 weeks)	Confirm Dynatrace Intelligence is generating problems correctly

```
// Post-migration validation — compare host count to your discovery inventory
fetch dt.entity.host
| summarize totalHosts = count()
| fieldsAdd target = "&lt;YOUR_DISCOVERY_COUNT&gt;", coverage = "Compare totalHosts to target"

// Smartscape equivalent (deprecated dt.entity.* still functional):
//   smartscapeNodes "HOST" | summarize totalHosts = count() | fieldsAdd target = "&lt;YOUR_DISCOVERY_COUNT&gt;", coverage = "..."
// Caveat: for validating against a discovery inventory keep the classic count — Smartscape's
// live-topology count omits hosts not currently in active topology.
```

```
// Post-migration validation — check for recent detected problems (confirms AI baseline is building)
fetch dt.davis.problems, from:-24h
| summarize problemCount = count(), by:{event.status}
| sort problemCount desc
```

7. Timeline Planning

Build your timeline working backward from the desired Managed decommission date.

Milestone Durations

Milestone	Typical Duration	Notes
Discover + Strategize + Design (Steps 1-3)	2-4 weeks	Front-load planning — rushed planning causes failed execution
Prepare (Step 4)	1-2 weeks	SaaS provisioning, ActiveGate deployment, network validation
Execute (Step 5, per wave)	1-2 weeks	Config migration + OneAgent redirect per wave
Integrate (Step 6)	1-2 weeks	Webhook, API, and ITSM reconnection
Parallel operation	2-4 weeks	Both environments running — critical for Dynatrace Intelligence baseline

Milestone	Typical Duration	Notes
Expand + Enable + Optimize (Steps 7-9)	2-4 weeks	New capabilities, tuning, adoption
Decommission	1 week	Shutdown Managed after final validation
Total typical	**4-12 weeks**	Depends on environment size and approach

Sample Timelines by Approach

Approach	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-12
Big Bang	Plan + Prepare	Execute + Validate	Parallel Run	Decommission	—
Phased by Env	Plan + Prepare	Dev wave	Staging wave	Prod wave	Parallel + Decommission
Phased by Region	Plan + Prepare	Region 1	Region 2	Region 3	Parallel + Decommission

> **Important:** The parallel operation period is non-negotiable. Dynatrace Intelligence needs 2-4 weeks to build baselines in SaaS before you can trust its problem detection. Do not decommission Managed until baselines are established.

Migration Plan Checklist

When building your high-level migration plan, ensure these items are addressed:

1. **Catalog and evaluate** monitoring components (OneAgents, AG/OA extensions, external sources) — any version updates needed?
2. **Identify** configurations and settings — what can be migrated automatically vs. manually?
3. **Leave legacy behind** — excessive or outdated configuration items can be left behind; apply naming standards

4. **Understand procedures and timeline** for AG, OA, and configuration migrations plus application restarts
5. **Minimize dual running** — provide a seamless cutover for Dynatrace users
   - What can be done over a weekend or outside of peak/business hours?
   - Group components into migration waves; learn from non-production first

Things to Include in Your Plan

Item	Detail
SSO implementation	IdP configuration, DNS domain validation, group mapping
Email allowlisting	Dynatrace email domain for SaaS platform/problem/security notifications
ActiveGate ordering	Procurement and provisioning of new AG machines
Root access requests	For new AG machines and monitored hosts
Network change requests	Connectivity between entities, AGs, proxies, and Dynatrace SaaS
Integration connectivity	External on-premise or SaaS systems, Dynatrace API, webhook integrations
Deployment automation	Prepare OA and AG deployment automation (Ansible, Puppet, Terraform, PowerShell)
Training	Train team on configuration migration tooling and SaaS administration
Execution plan	Environment configurations and application configurations, in what order

8. The 90/10 Rule

Based on hundreds of successful migrations, a consistent pattern emerges:

> **90% of configurations migrate automatically** via the SaaS Upgrade Assistant. The remaining **10% takes 90% of the manual effort**.

What Migrates Automatically (the 90%)

Category	Tool
Settings and configuration	SaaS Upgrade Assistant
Dashboards	SaaS Upgrade Assistant
Alert policies	SaaS Upgrade Assistant
Management zones	SaaS Upgrade Assistant
Auto-tag rules	SaaS Upgrade Assistant
Request attributes	SaaS Upgrade Assistant
Calculated metrics	SaaS Upgrade Assistant

What Requires Manual Effort (the 10%)

Item	Why Manual	Effort Level
Credential Vault entries	Secrets cannot be exported from Managed	High — must recreate each credential
Problem notification webhooks	URLs and tokens change between environments	Medium — update each endpoint
Cloud platform credentials	AWS/Azure/GCP keys are tenant-specific	Medium — reconfigure each integration
Synthetic private locations	Tied to Managed ActiveGates	Medium — recreate with SaaS ActiveGates
Custom extensions (v1)	Legacy format not supported in SaaS	High — convert to Extensions 2.0
Custom scripts and automation	Hardcoded Managed URLs and tokens	Medium — update all references
Third-party ITSM integrations	Webhook endpoints change	Medium — reconfigure each connection

Budget Accordingly

Planning Item	Recommendation
Identify the 10% early	Use your discovery inventory to list every manual item
Assign owners	Each manual item needs a person responsible
Track separately	Manual items need their own checklist — do not mix with automated migration
Test before cutover	Every manual item should be validated in SaaS before redirecting agents

9. Step Completion Checklist

Do not proceed to Step 3 (Design) until all items are confirmed.

Checkpoint	Status
Migration approach selected (Big Bang / Phased by Env / Phased by Region)	[ ]
Order of operations documented and reviewed with stakeholders	[ ]
Risk assessment completed with owners assigned to each risk	[ ]
Success criteria defined with measurable targets	[ ]
High-level timeline established with milestone dates	[ ]
Dynatrace account team engaged for licensing and migration support	[ ]
90/10 manual effort items identified and owners assigned	[ ]
Communication plan drafted for affected teams	[ ]
Rollback procedure documented	[ ]

Next Step

> **M2S-03: Step 3 — Design** — Create your target SaaS architecture, including network topology, ActiveGate placement, IAM model, and configuration mapping.

Summary

In this notebook, you:

- Reviewed the three-phase, nine-step migration framework
- Selected a migration approach based on your environment size and risk tolerance
- Documented the required order of operations and critical dependencies
- Assessed key considerations across licensing, data, infrastructure, configuration, and security
- Completed a risk register with mitigations and owners
- Defined measurable success criteria for migration completion
- Built a timeline with milestone durations
- Identified the 90/10 split between automated and manual migration effort

> **Key Takeaway:** Strategy is about making decisions before the pressure is on. A clear approach, documented risks, and measurable success criteria are what separate migrations that succeed from those that spiral into firefighting.

Continue to M2S-03: Step 3 — Design to create your target SaaS architecture.
